

**Department of Electronics and Telecommunication Engineering
(NBA ACCREDITED)
Digital System Design Laboratory
Academic Year 2025-2026, SEM-5, Odd Semester**

Course Code	EC24P
Subject Professor In-charge	Prof. Satendra Mane
Student Name	Rahul Bhadane
Roll Number	23104B0003
Class	T.E – EXTC/EXCS SEM-V
Division	B
Date of Performance	5 th September 2025
Date of Submission	12 th September 2025

EXPERIMENT NO.5

Estimated time to complete this experiment: 02 hours.

Objective: To design and implement T flipflop

CO to be achieved: CO1, CO2

Expected Outcome of Experiment:

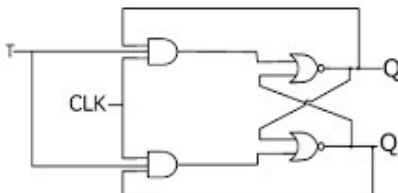
- Students will learn to design, simulate and synthesis the arithmetic circuit using VIVADO & VERILOG.

Pre Lab/ Prior Concept: Designing of FlipFlops.

Apparatus/TOOL:

- PC
- VIVADO TOOL
- SPARTAN-7 FPGA Board.

CIRCUIT DIAGRAM.



VERILOG -CODE.

```
module t_flipflop (
    input wire clk, // Clock input
    input wire rst, // Asynchronous reset
    input wire t,   // Toggle input
    output reg q    // Output
);
```

always @(posedge clk or posedge rst) begin

if (rst)

q <= 1'b0; // Reset output to 0

else if (t)

q <= ~q; // Toggle output when T=1

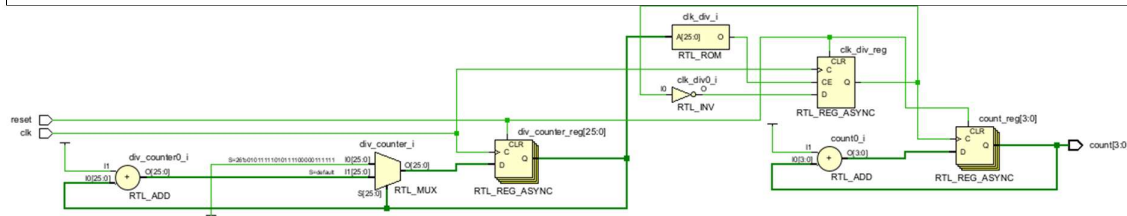
else

q <= q; // Hold previous state when T=0

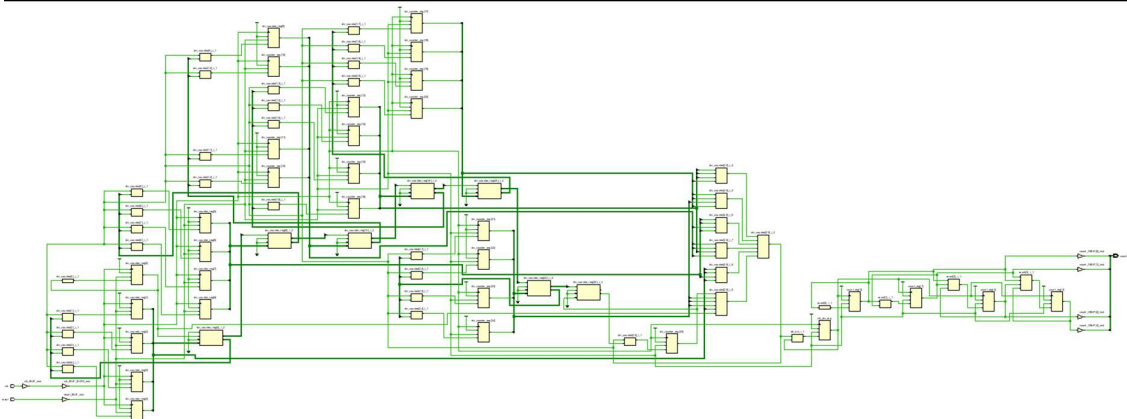
end

endmodule

SIMULATION RESULTS.



SYNTHESIS MODEL.



CONCLUSION.

The 4-bit ALU was successfully designed and implemented using Verilog in Vivado. It performs arithmetic (addition, subtraction, increment, decrement) and logical (AND, OR, XOR, NOT) operations correctly. The simulation verified its functionality, and the experiment helped understand ALU design, simulation, and synthesis on FPGA

ASSESSMENT RECORD.

Practical Performance [Model/Code, Results] (5 Marks)	Presentation/ Learning Outcome [Observation, Conclusion] (5 Marks)	Total (10 Marks)	Sign
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